R21

Code No.: U21PC602CS

MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE

(Autonomous)

B.E. VI Semester (Main) (Branch: CSE) Examination, July 2024

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours Max. Marks : 70

Note : i) FIRST Question is compulsory and answer any **FOUR** questions from the remaining six questions. Each question carries 14 Marks.
ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.
iii) Missing data, if any, may suitably be assumed.

Q.No.			Marks	CO	BT
1.	a)	Define artificial intelligence.	(2)	1	L1
	b)	Compare parametric and non-parametric machine learning models.	(2)	2	L2
	c)	Mention the objective of logistic regression.	(2)	3	L2
	d)	How to calculate precision, recall and accuracy?	(2)	4	L2
	e)	List out the applications of reinforcement learning.	(2)	5	L1
	f)	Differentiate informed and uninformed search strategies.	(2)	1	L2
	g)	Distinguish between Agglomerative and divisive clustering.	(2)	5	L4
2	(a)	Compare and contrast different types of agents.	(7)	1	L2
	(b)	Analyze the procedure for implementation of the best-first search.	(7)	1	L4
3	(a)	Elaborate various phases in building the expert systems.	(7)	2	L2
	(b)	Describe the resolution method for theorem proving in propositional logic.	(7)	2	L2
4	(a)	Explain the different types of machine learning algorithms.	(7)	3	L2
	(b)	Examine the methodology followed by naïve bayes classification.	(7)	3	L4
5	(a)	Show the mechanism of a multilayer neural network.	(7)	4	L2
	(b)	Diagnose the role of confusion matrix for model evaluation	(7)	4	L4
6	(a)	Illustrate K- means algorithm with example.	(7)	5	L2
	(b)	Compare policy vs reward and active and passive reinforcement learning.	(7)	5	L4
7	(a)	Elucidate the syntax and semantics of first-order logic.	(5)	2	L2
	(b)	Infer the functionality of K-nearest neighbors' algorithm.	(5)	3	L4
	(c)	Discuss the importance of various activation functions in neural networks.	(4)	5	L2

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